

a.sort(), sorted(a), np.argsort(a) and np.lexsort(b, a) in Python

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These are all different types for sorting techniques that behave very differently. Let's study which technique works how and which one to use.

Let 'a' be a *numpy* array

- **a.sort()**

- (i) Sorts the array in-place & returns None
- (ii) Return type is None
- (iii) Occupies less space. No copy created as it directly sorts the original array
- (iv) Faster than sorted(a)

```
# Python code to sort an array in-place
# using a.sort
import numpy as np

# Numpy array created
a = np.array([9, 3, 1, 7, 4, 3, 6])

# unsorted array print
print('Original array:\n', a)

# Return type is None
print('Return type:', a.sort())

# Sorted array output
print('Original array sorted->', a)
```

```
OUTPUT: For a.sort()
Original array:
[9 3 1 7 4 3 6]
Return type: None
Original array sorted-> [1 3 3 4 6 7 9]
```

- **sorted(a)**

- (i) Creates a new list from the old & returns the new one, sorted
- (ii) Return type is a list
- (iii) Occupies more space as copy of original array is created and then sorting is done
- (iv) Slower than a.sort()



```
# Numpy array created
a = np.array([9, 3, 1, 7, 4, 3, 6])

# unsorted array print
print('Original array:\n', a)
b = sorted(a)

# sorted list returned to b, b type is
# <class 'list'>
print('New array sorted->', b)

# original array no change
print('Original array->', a)
```

```
OUTPUT:a.sorted()
Original array:
[9 3 1 7 4 3 6]
New array sorted-> [1, 3, 3, 4, 6, 7, 9]
Original array-> [9 3 1 7 4 3 6]
```

- **np.argsort(a)**

- (i) Returns the indices that would sort an array
- (ii) Return type is numpy array
- (iii) Occupies space as a new array of sorted indices is returned

```
# Python code to demonstrate working of np.argsort
import numpy as np

# Numpy array created
a = np.array([9, 3, 1, 7, 4, 3, 6])

# unsorted array print
print('Original array:\n', a)
```

```
# To get sorted array using sorted indices
# c is temp array created of same len as of b
c = np.zeros(len(b), dtype = int)
for i in range(0, len(b)):
    c[i] = a[b[i]]
print('Sorted array->', c)
```

```
OUTPUT:np.argsort(a)
Original array:
[9 3 1 7 4 3 6]
Sorted indices of original array-> [2 1 5 4 6 3 0]
Sorted array-> [1 3 3 4 6 7 9]
```

• np.lexsort((b, a))

- (i) Perform an indirect sort using a sequence of keys
- (ii) Sort by a, then by b
- (iii) Return type ndarray of ints Array of indices that sort the keys along the specified axis
- (iv) Occupies space as a new array of sorted indices pair wise is returned.

```
# Python code to demonstrate working of
# np.lexsort()
import numpy as np

# Numpy array created
a = np.array([9, 3, 1, 3, 4, 3, 6]) # First column
b = np.array([4, 6, 9, 2, 1, 8, 7]) # Second column
print('column a, column b')
for (i, j) in zip(a, b):
    print(i, ' ', j)

ind = np.lexsort((b, a)) # Sort by a then by b
print('Sorted indices->', ind)
```

```
OUTPUT:np.lexsort((b, a))
column a, column b
9    4
3    6
1    9
3    2
4    1
3    8
6    7
Sorted indices-> [2 3 1 5 4 6 0]
```



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